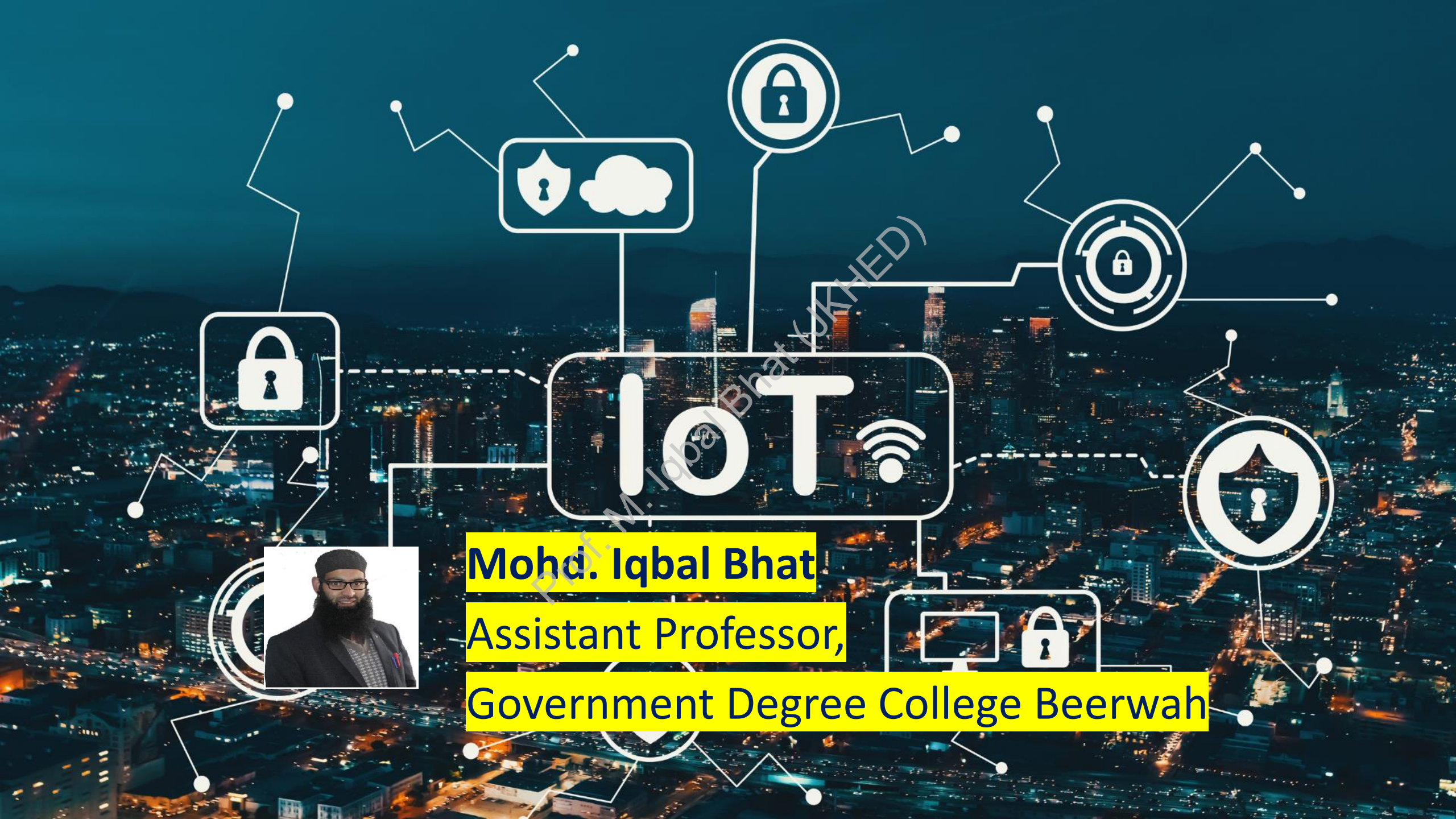


A graphic illustration centered around IoT security. A large, rounded rectangle in the center contains the text 'IoT' in a bold, white, sans-serif font, followed by a white Wi-Fi symbol. This central box is connected by white lines to several other icons: a padlock inside a circle at the top, a shield and cloud inside a rounded rectangle at the top left, a target with a padlock inside a circle at the top right, a shield with a padlock inside a circle on the right, and a monitor with a padlock inside a rounded rectangle at the bottom right. On the left, there is a smaller rounded rectangle with a padlock icon. The background is a dark, blue-toned aerial view of a city at night, with lights from buildings and streets visible. A diagonal watermark text 'Mohd. Iqbal Bhat (UKHED)' is overlaid across the center of the image.

IoT



Mohd. Iqbal Bhat

Assistant Professor,

Government Degree College Beerwah

Syllabus

UNIT - 1 (15 Hours)

Introduction: Definition & Characteristics of Iot, Physical Design of Iot, Basic Components of IoT, Thing, Sensing & Actuators, Vision, Physical Parameters. Iot Protocols.

Application Areas of IoT: Home Automation, Smart Cities, Medical, Logistics, Environment, Analytics. Smart Grids.

UNIT - 2 (15 Hours)

Iot Communication Models, APIs, IoT Architecture: Basic Architecture: 3 layer and 5-layer Architecture, ITU-IoT Reference Model, Machine to Machine Communication, IoT Gateways, Wireless Sensor Networks. Technologies: Bluetooth Low Energy (BLE), ZigBee: Architecture, Comparison with other wireless standards. LoraWAN.

UNIT - 3 (15 Hours)

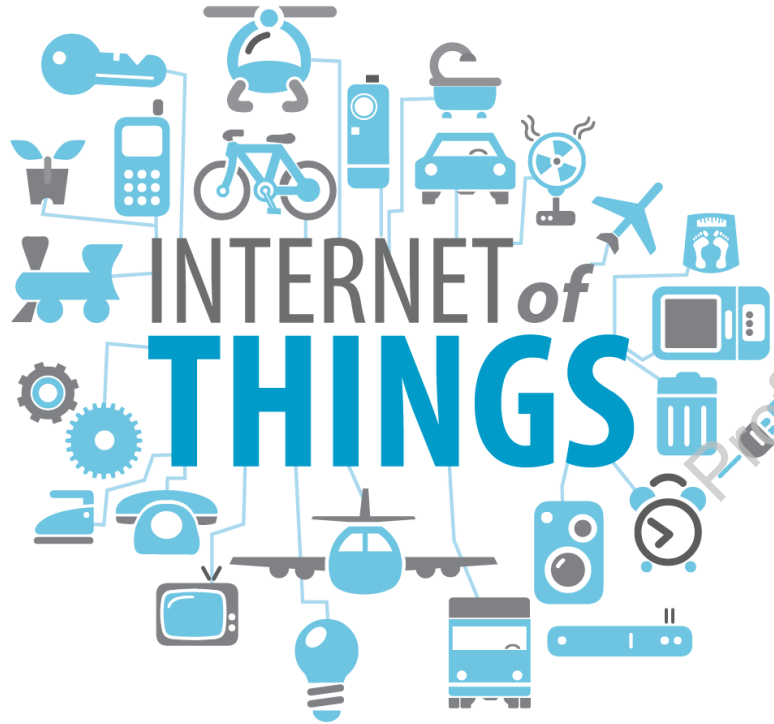
Electronic Product Code (EPC), Near Field Comm. (NFC), 6LoWPAN, End to End Reliability: MQTT, SCADA. Hardware and Software Platforms: Hardware: Raspberry Pi, ESP8266 Wifi Module, Arduino. Implementation of Basic Sensors (temperature, humidity, proximity, gas, air quality, Ultrasonic sensors)

Internet of Things Privacy and Security Issues, Steps towards a security platform in IoT

Text Books:

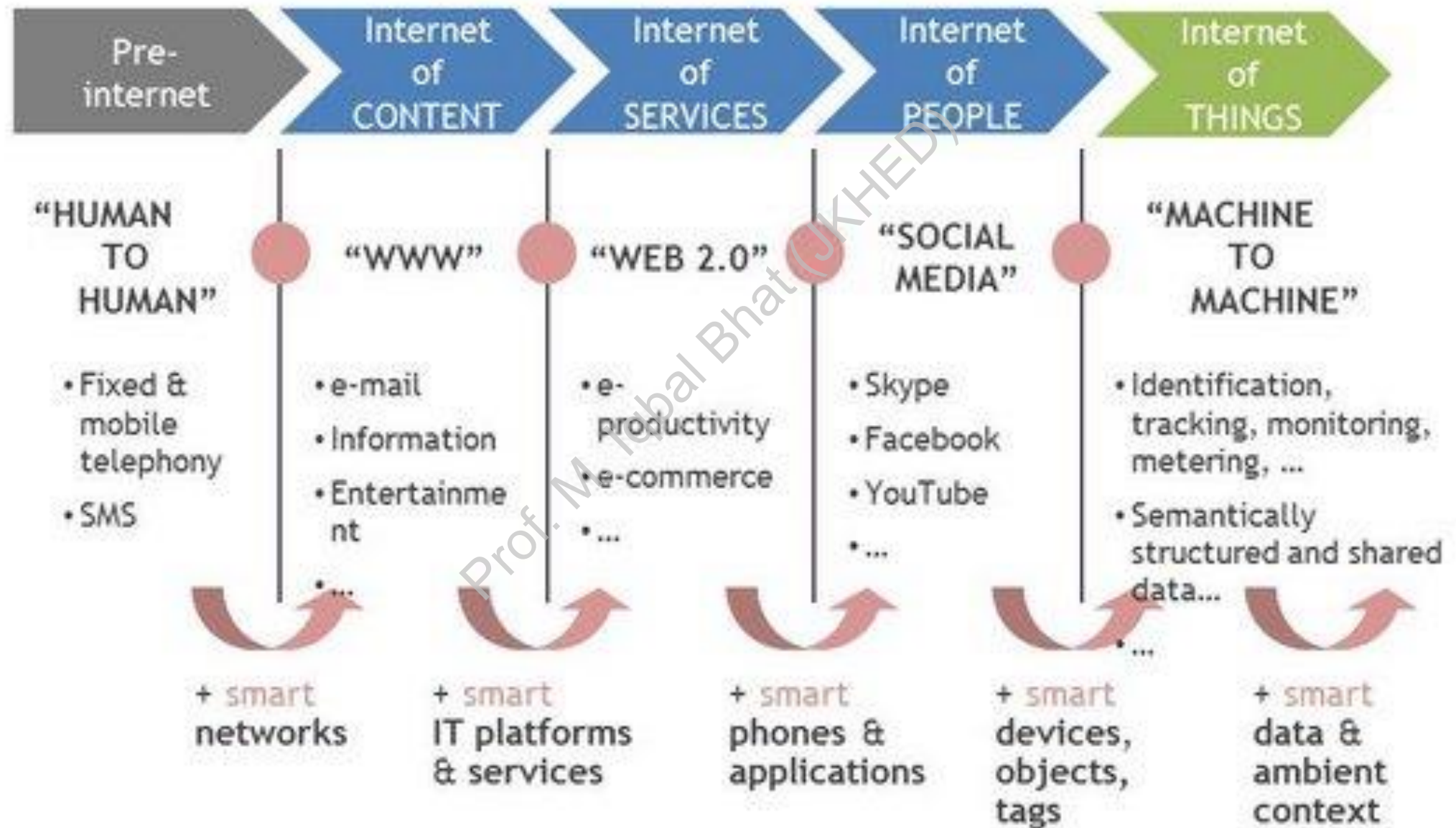
1. Vijay Madisetti and Arshdeep Bahga, "Internet of Things (A Hands-on-Approach)", 1st Edition, VPT, 2014. (ISBN-13: 978-8173719547)
2. Internet of Things (IoT), Dr. Kamlesh Lakhwani, Dr. Hemant Kumar Gianey, Joseph Kofi Wireko, Kamal Kant Hiran

What is



IoT Internet
of
Things

Evolution of IoT



IoT is.... (According to EU)



- A global network infrastructure, linking physical and virtual objects through the exploitation of data capture and communication capabilities

IoT is.... (According to ITU)



A global infrastructure for the information society, enabling advanced services by interconnecting (physical and virtual) things based on, existing and evolving, interoperable information and communication technologies [ITU-T Y.2060]

IoT is.... (According to IETF)



IoT is a world-wide network of interconnected objects uniquely addressable, based on standard communication protocols

IoT is.... (According to IEEE)



A network of items each
embedded with sensors which
are connected to the Internet.



IoT is.... (According to Wikipedia)

The Internet of Things (IoT) is the network of physical devices, vehicles, home appliances, and other items embedded with electronics, software, sensors, actuators, and connectivity which enables these things to connect and exchange data, creating opportunities for more direct integration of the physical world into computer-based systems, resulting in efficiency improvements, economic benefits, and reduced human exertions.



*The **Internet of things (IoT)** is the network of physical devices, vehicles, home appliances and other items embedded with **electronics, software, sensors, actuators, and network connectivity** which enables these objects to connect and exchange data.*

IoT Inter of Things



A dynamic global network infrastructure with self-configuring capabilities based on standard and interoperable communication protocols where physical and virtual ‘things’ have identities, physical attributes, and virtual personalities and use intelligent interfaces, and are seamlessly integrated into the information network, often communicate data associated with users and their environments.”

IoT

Dynamic & Self-Adapting: IoT devices and systems may have the capability to dynamically adapt with the changing contexts and take actions based on their operating conditions, user's context, or sensed environment.

Self-Configuring: Devices autonomously configure for specific functionalities.

Interoperable Protocols: Supports various communication standards for compatibility.

Unique Identities: Physical and virtual "things" with distinct identities and attributes.

Intelligent Interfaces: Seamless integration into the information network for smart interactions.

Data Communication: Shares data related to users and their environments.



Characteristics of IoT

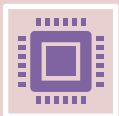
Characteristics of IoT



Interconnectivity: IoT networks connect diverse devices, ensuring data compatibility and accessibility through embedded sensors.



Intelligence: Devices in IoT networks are self-aware and make context-based decisions, sharing and accessing data among themselves.



Dynamic Nature: IoT systems adapt to changes, such as device states and network configurations, making them suitable for dynamic environments.



Sensing: Continuous monitoring through sensors detects environmental changes, aiding in device interconnectivity and smart city applications.

Characteristics of IoT



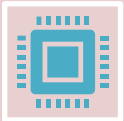
Expressing: IoT systems communicate with end-users through GUIs or direct actions, like service notifications for maintenance.



Extensive Scale: IoT networks manage vast numbers of devices across large geographic areas, with an estimated 75 billion devices by 2025.



Heterogeneity: IoT supports diverse hardware and networks, treating all devices as if they share common properties for seamless integration.



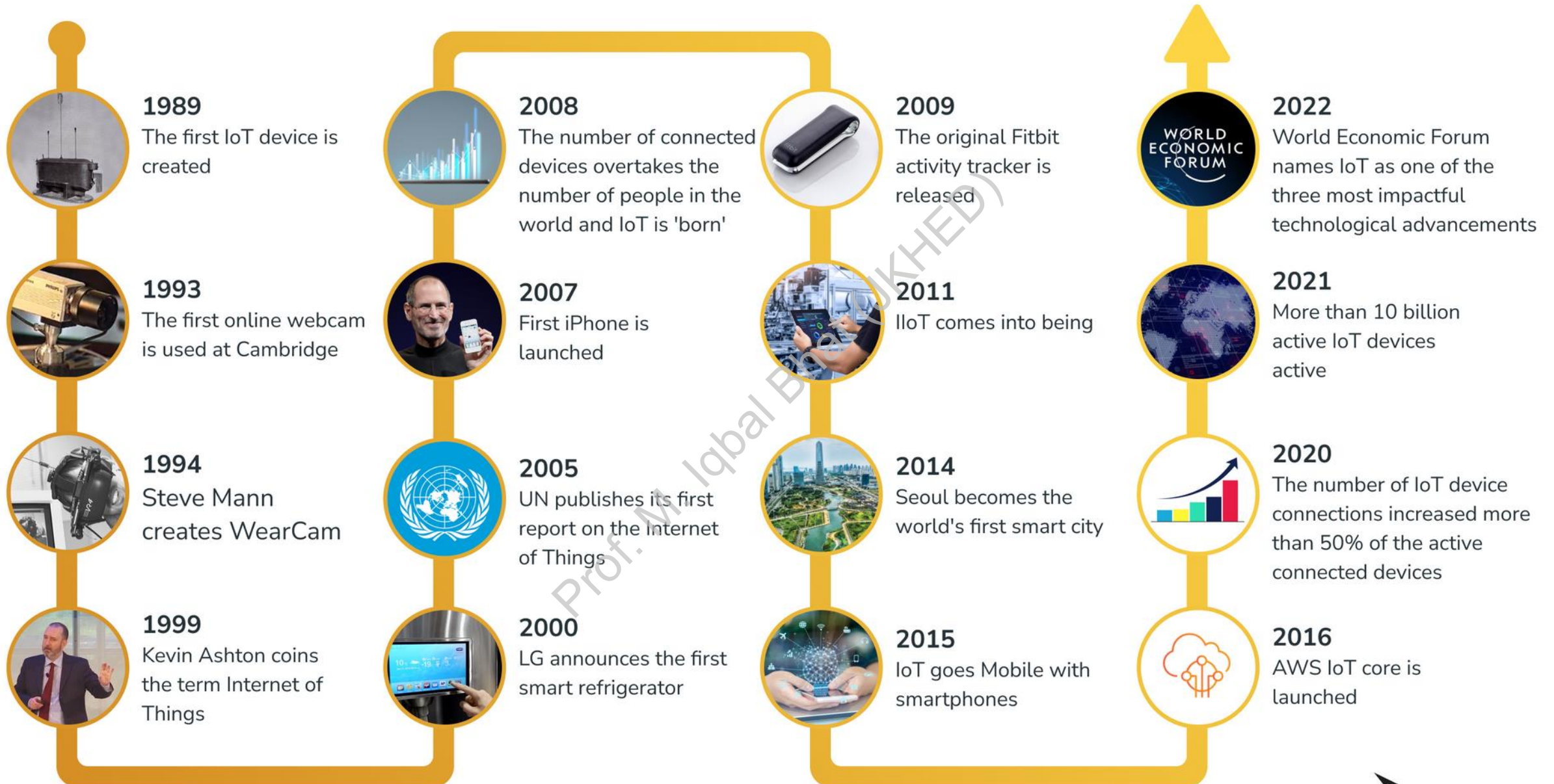
Endpoint Management: Essential for system integrity, endpoint management ensures the proper functioning of IoT applications, like automated ordering in smart devices.



Security: IoT devices implement security measures like encryption and self-monitoring sensors to protect endpoints, data, and physical integrity while maintaining transparency.

History of IoT





Source: [Evolution of Internet of Things \(IoT\) - A Brief History \(bytebeam.io\)](https://bytebeam.io/evolution-of-internet-of-things-iot-a-brief-history/)

History of IoT



Wearable computer with eyeglass mounted display to aid in lip reading



First wearable computer inside a shoe

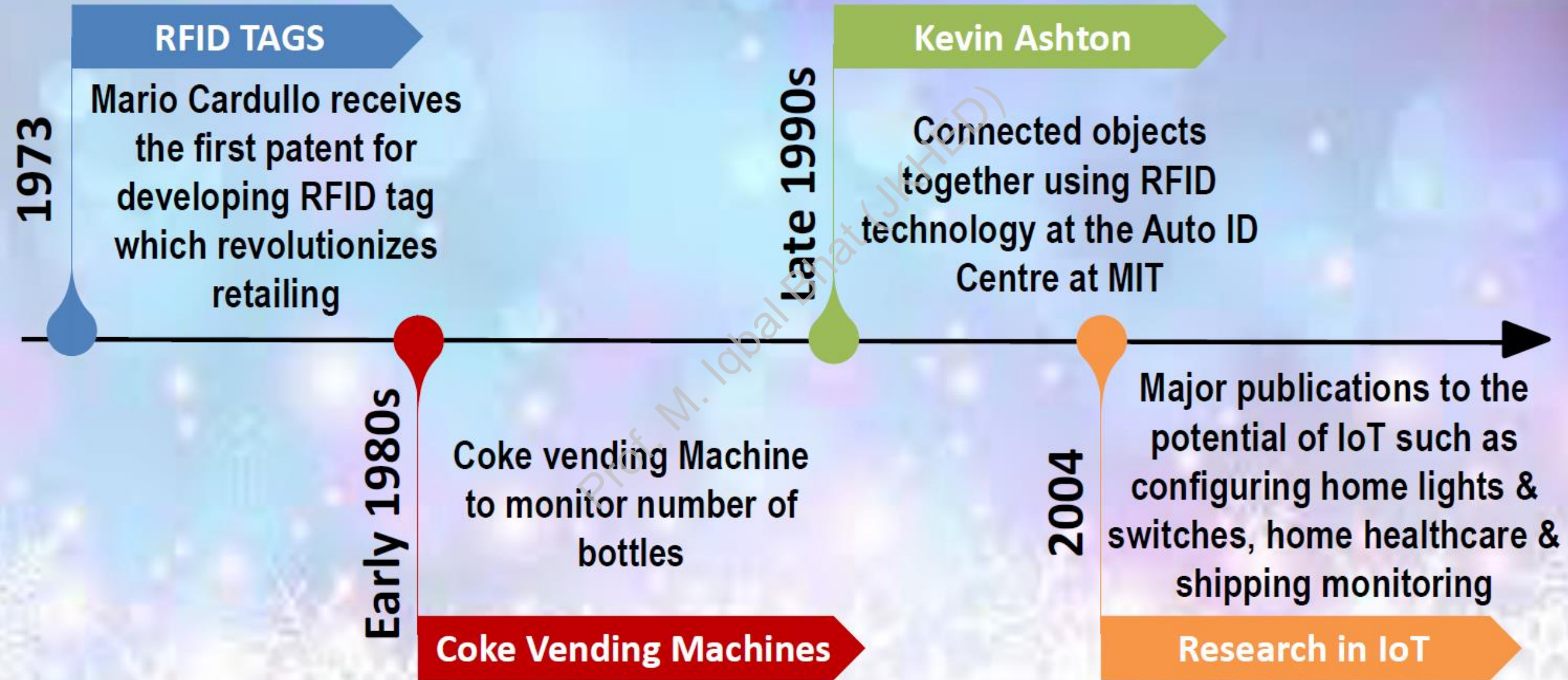


Head Mounted Display



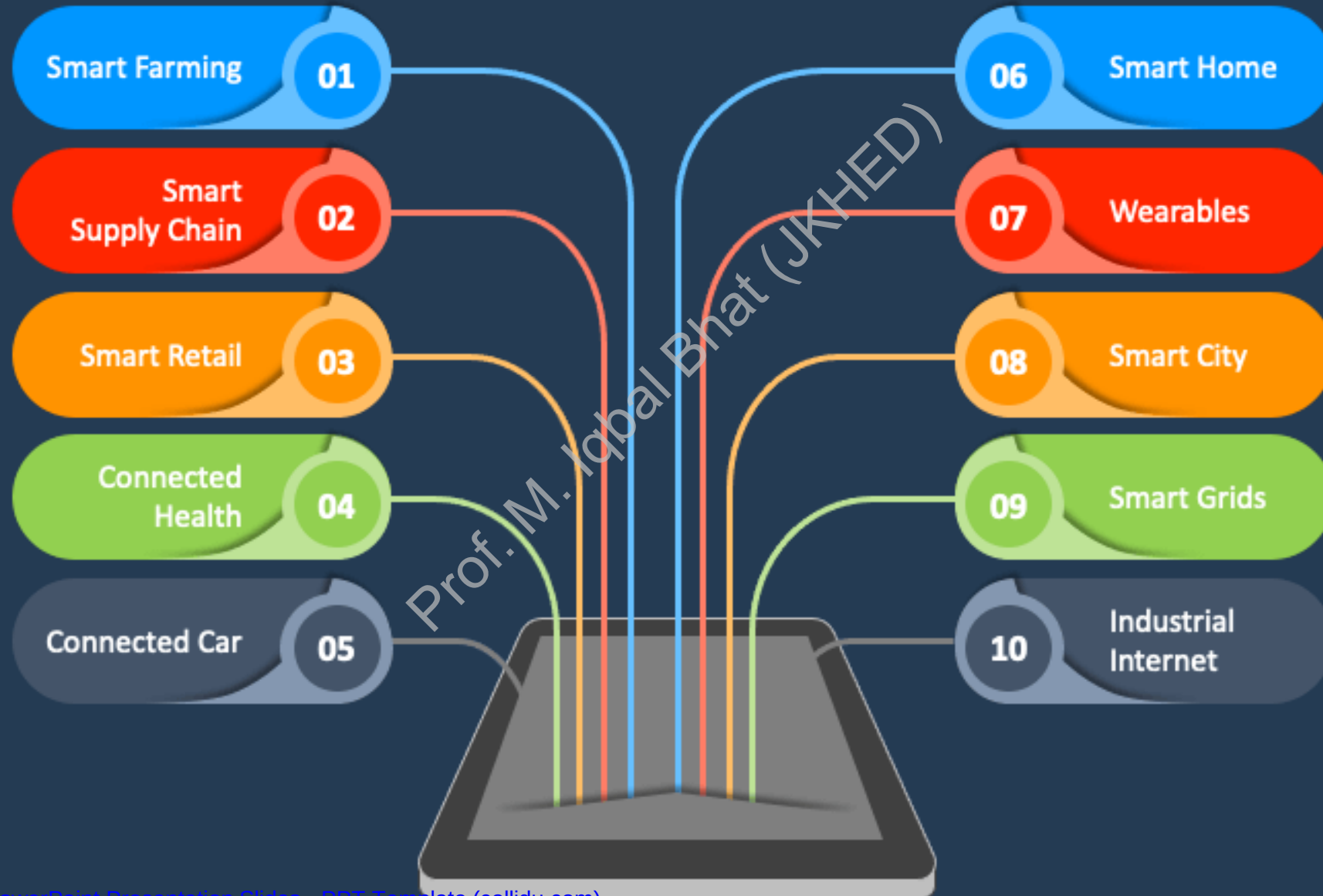
1945: Norman Joseph Woodland invented Bar code. He obtained his idea when he drew four lines in the sand at a beach in Miami

History of IoT



APPLICATIONS OF IoT

IoT Applications and Use Cases





What the Internet of Things Means for **Education**

Solutions IoT has in education



Communication
network



New generation of
textbooks



Data collection



Advanced security



Advanced school
management

prof. M. Iqbal Bhatti (JKHED)

Advantages of IoT for education actors



Student

- Adaptable learning
- Smart software for learning
- Improved learning results
- Adaptable learning



Instructor

- Smart tools for teaching
- Attendance tracking
- Flexibility with education process



Administration

- Smart monitoring of the building
- Higher security
- Modified LMS

Cyber School with IOT 21st Century School

Smart School Transportation



Smart School Office

1



Student IoT Project Management



Smart School Building Management



Smart Student Health



Smart Classrooms

5



Smart STEAM Lab

6



Smart Cafeteria/Minimart



Student Activity Tracking

8





IOT and Healthcare





Asset
Monitoring



Triggers
/alerts

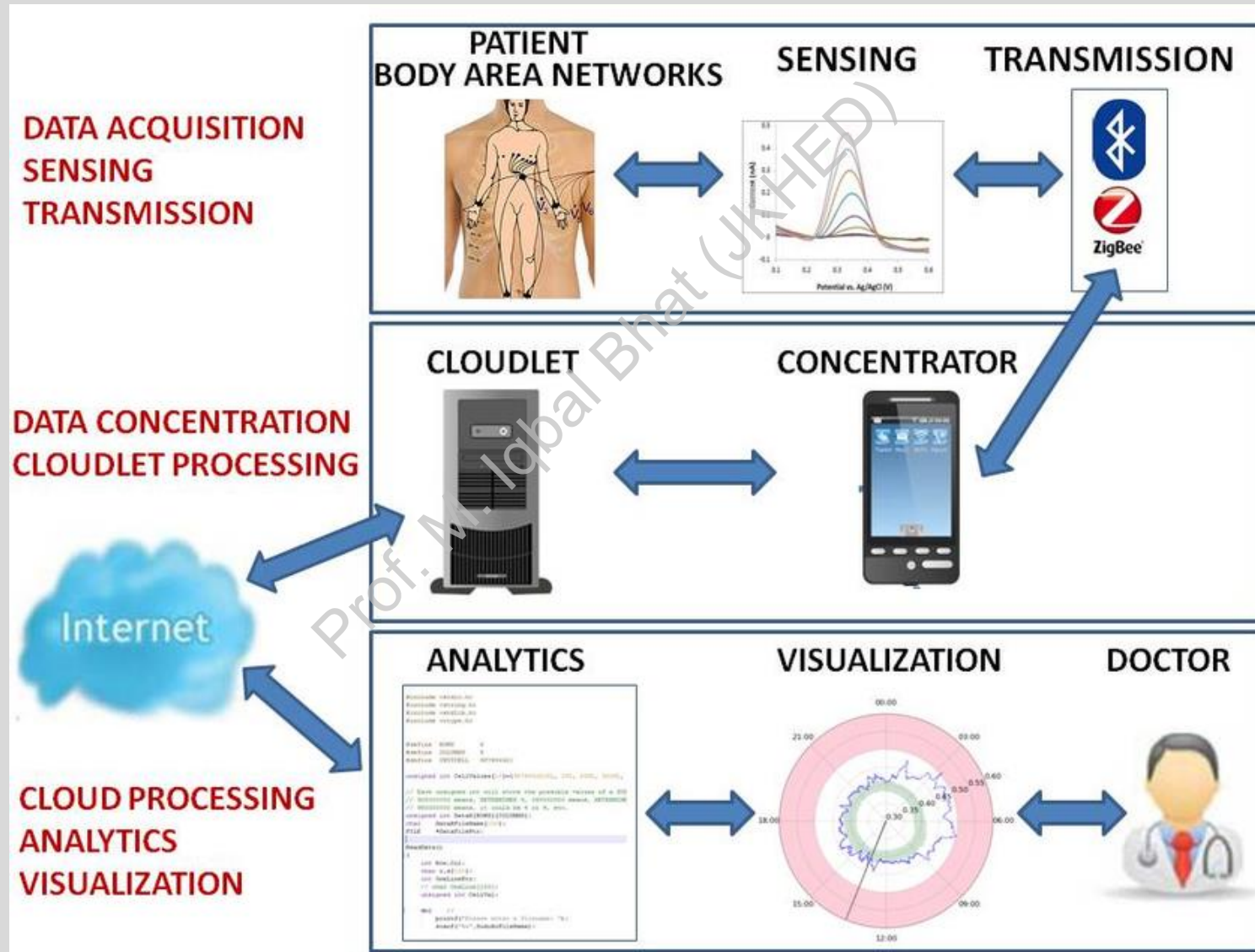


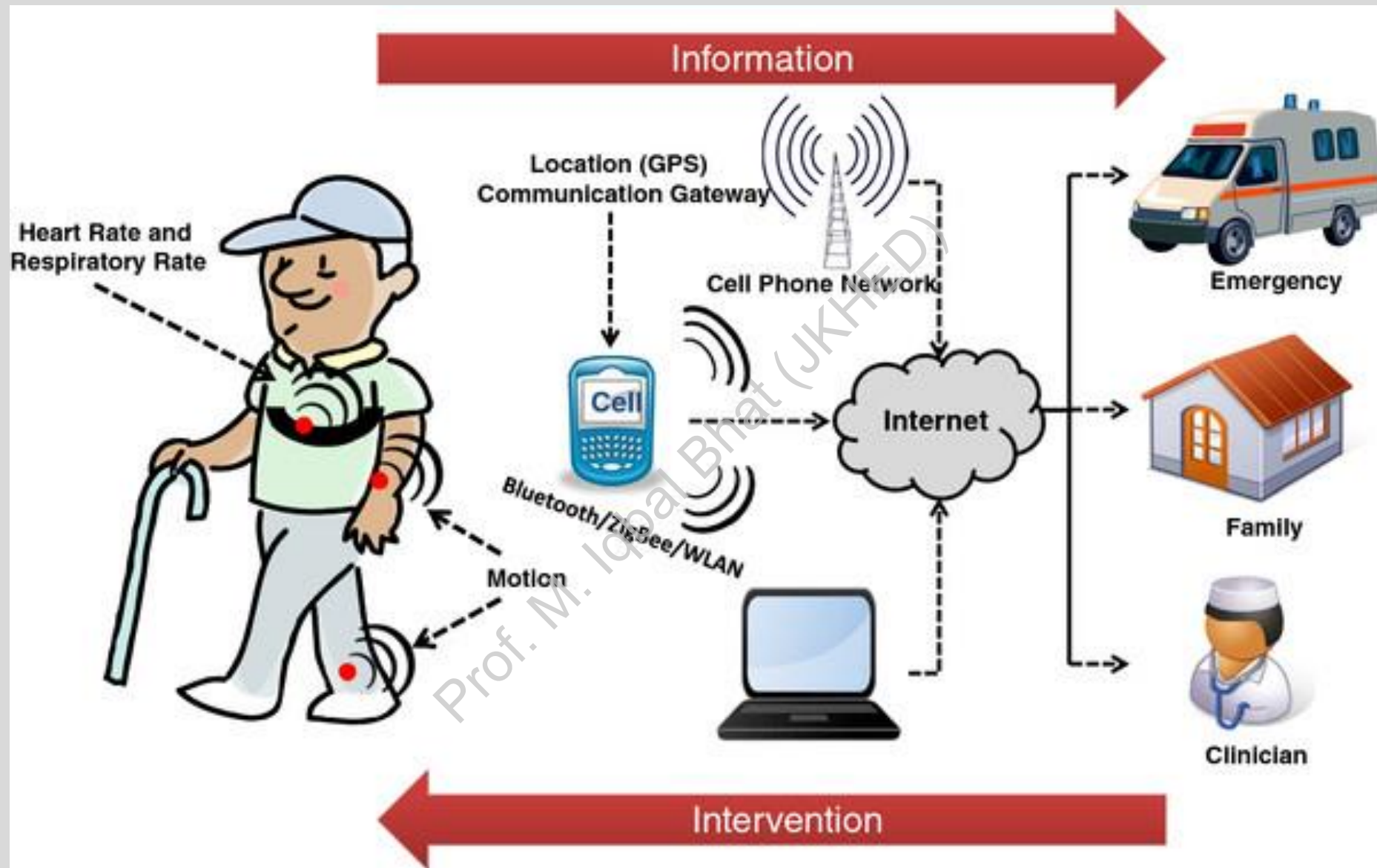
Data analysis

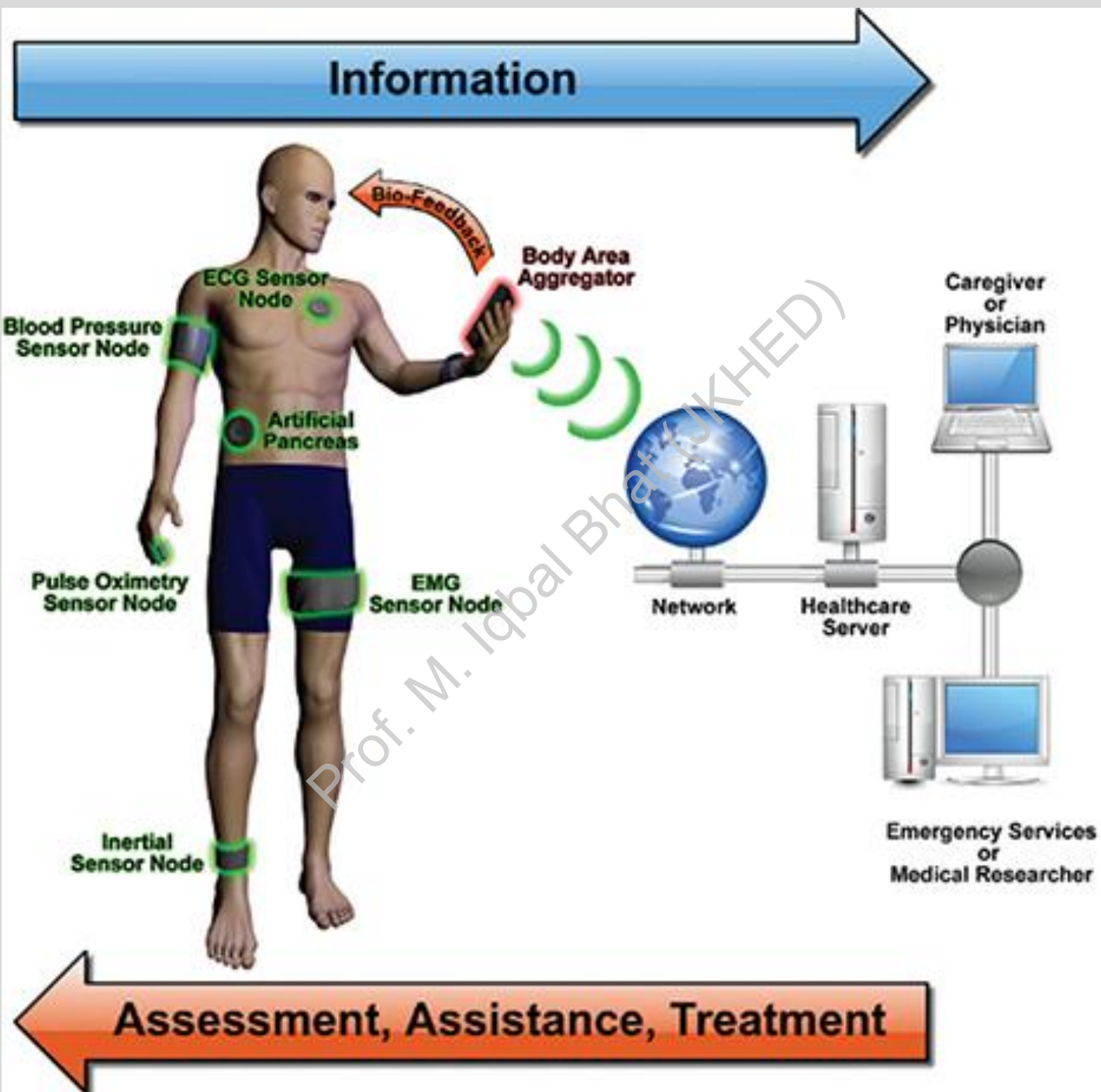


Prof. M. Iqbal Bhat (JKHED)

Adjustable Patient Monitoring



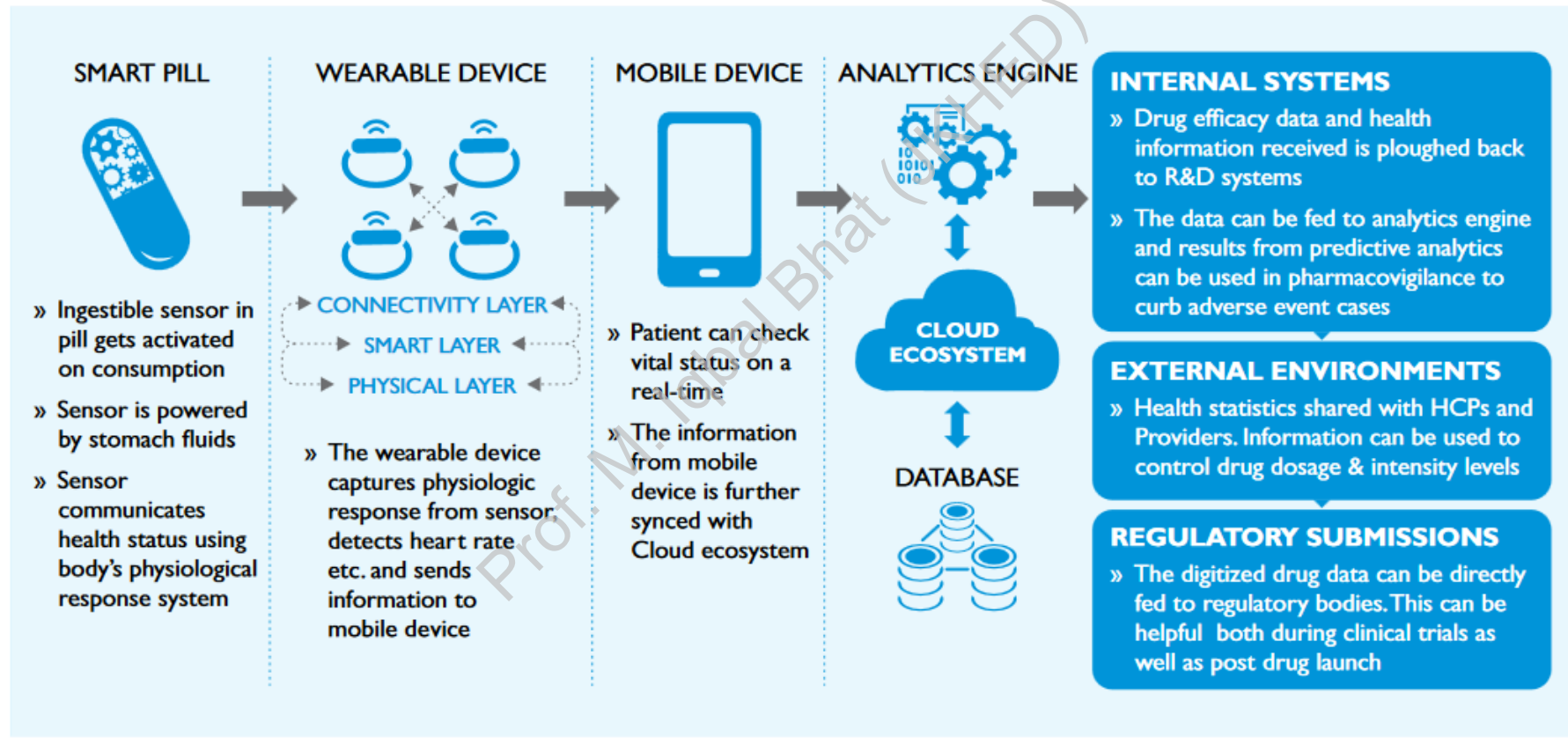




IOT Based Monitoring System



Enhanced Drug Management



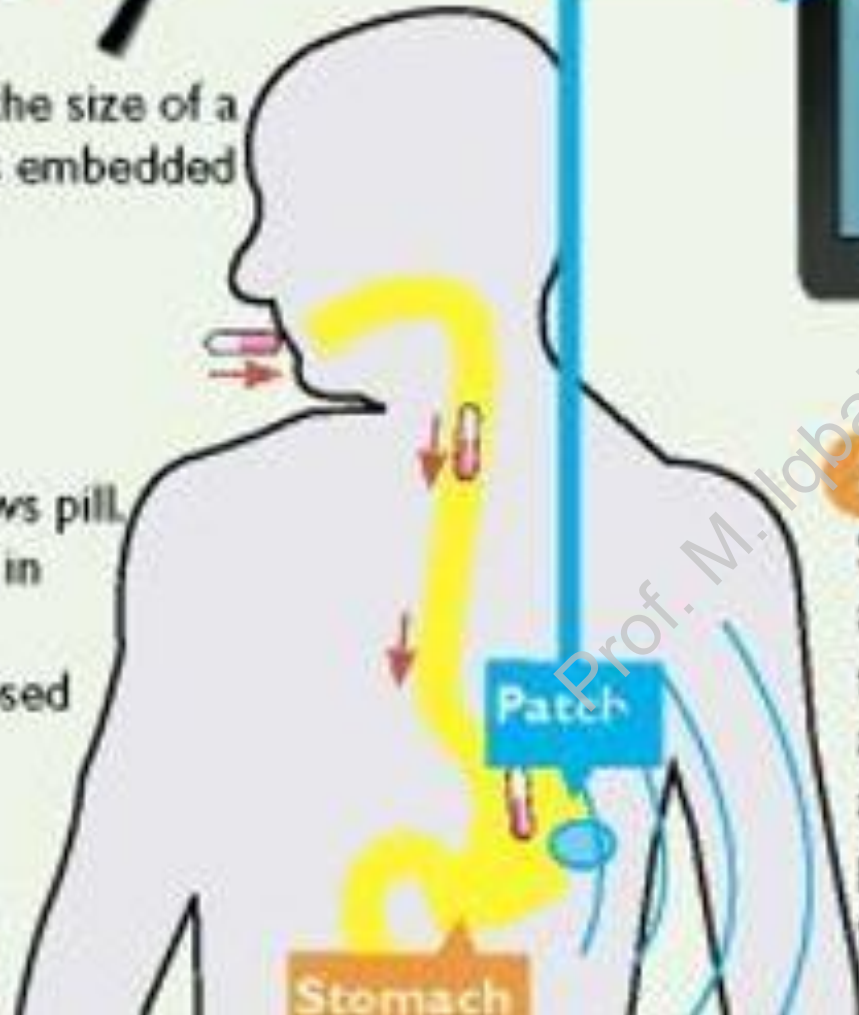
DRUGS GO DIGITAL



A silicon chip the size of a grain of sand is embedded in the pill

1

Patient swallows pill. As it dissolves in stomach acid, sensor is exposed



2

Sensor is activated by stomach acid and transmits a signal to patch on stomach

3

The patch sends the data via Bluetooth



4

The mobile phone automatically sends a message to doctor or carer



Available to buy or in development today...



ELECTRONIC PILL DISPENSERS

Designed to remind individuals to take their medication at the right time and to ease the burden of complex medication regimens, electronic pill dispensers such as my uBox and MedMinder alert both the patient and their caregivers.



ELECTRONIC BOTTLES, CAPS AND POUCHES

Wireless smart pill bottles, such as Adheretech, measure the volume of tablets or liquid left in a bottle, while GlowCaps use light and sound to signal when it's time to take your medication. Inhaler attachments, such as GeckoCap and Asthmapolis, monitor where and when an inhaler is used.



'PHARMACY ON A CHIP'

Currently undergoing clinical trials, microchips drug delivery technologies administer controlled doses of a drug at precisely the right time via a microchip inserted on the waist. Still in its infancy, the technology holds promise for improved patient adherence managed remotely.



BIOMONITORING DRUGS

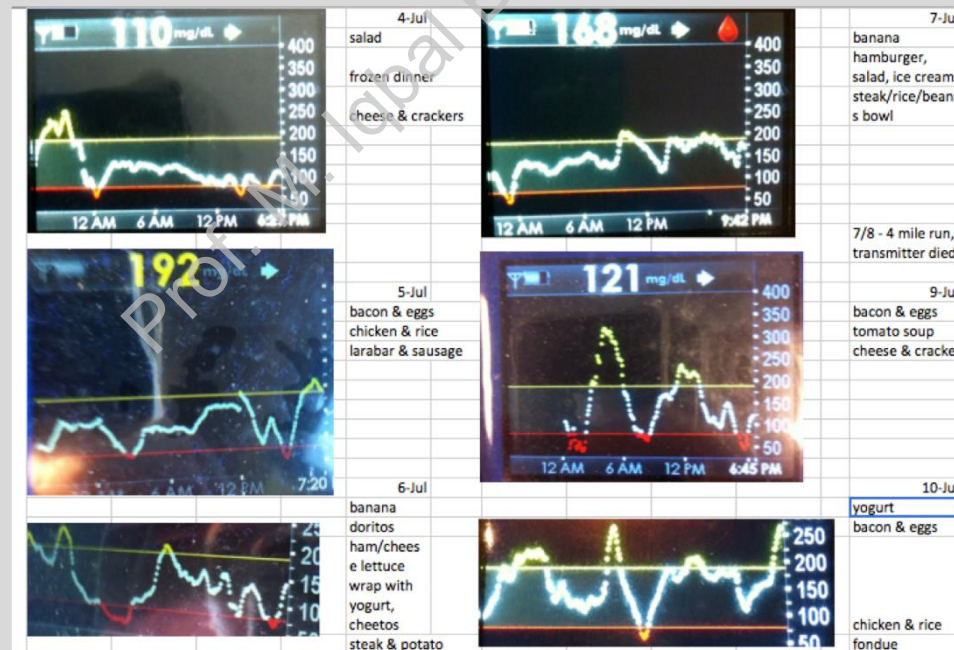
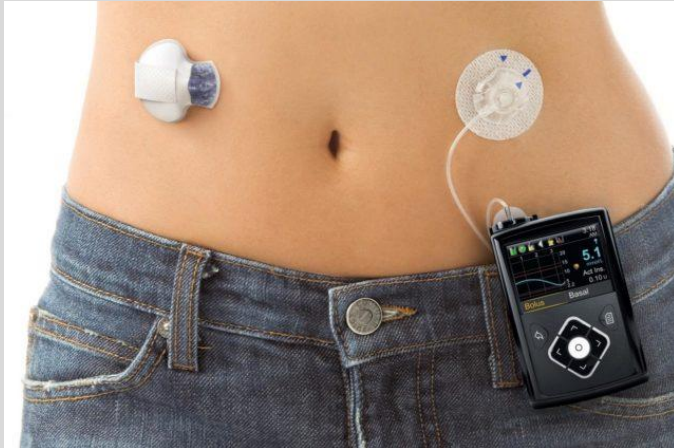
Ingestible sensors as small as a grain of sand exist today. Helius by Proteus Digital Health is a digital health feedback system. Embedded in a tablet, sensors communicate with a patch worn on the stomach. This then relays information to your phone, and further to your support network and care providers.



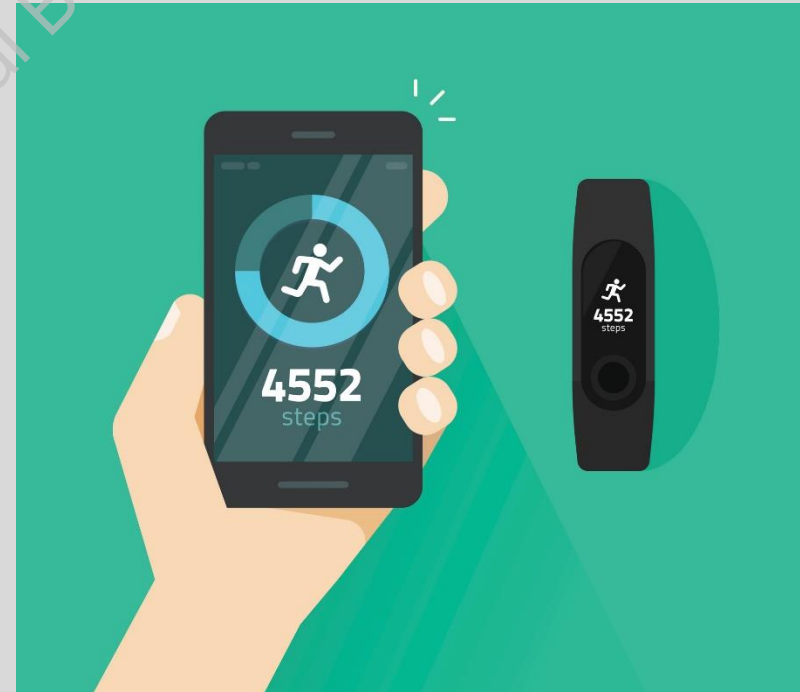
SMART WATCHES

Already designed to act as a health and fitness companion with all the capabilities of a fitness tracker, smart watches (such as Apple Watch) have the potential to integrate with multiple technologies, including those described here.

OpenAPS - closed-loop insulin delivery



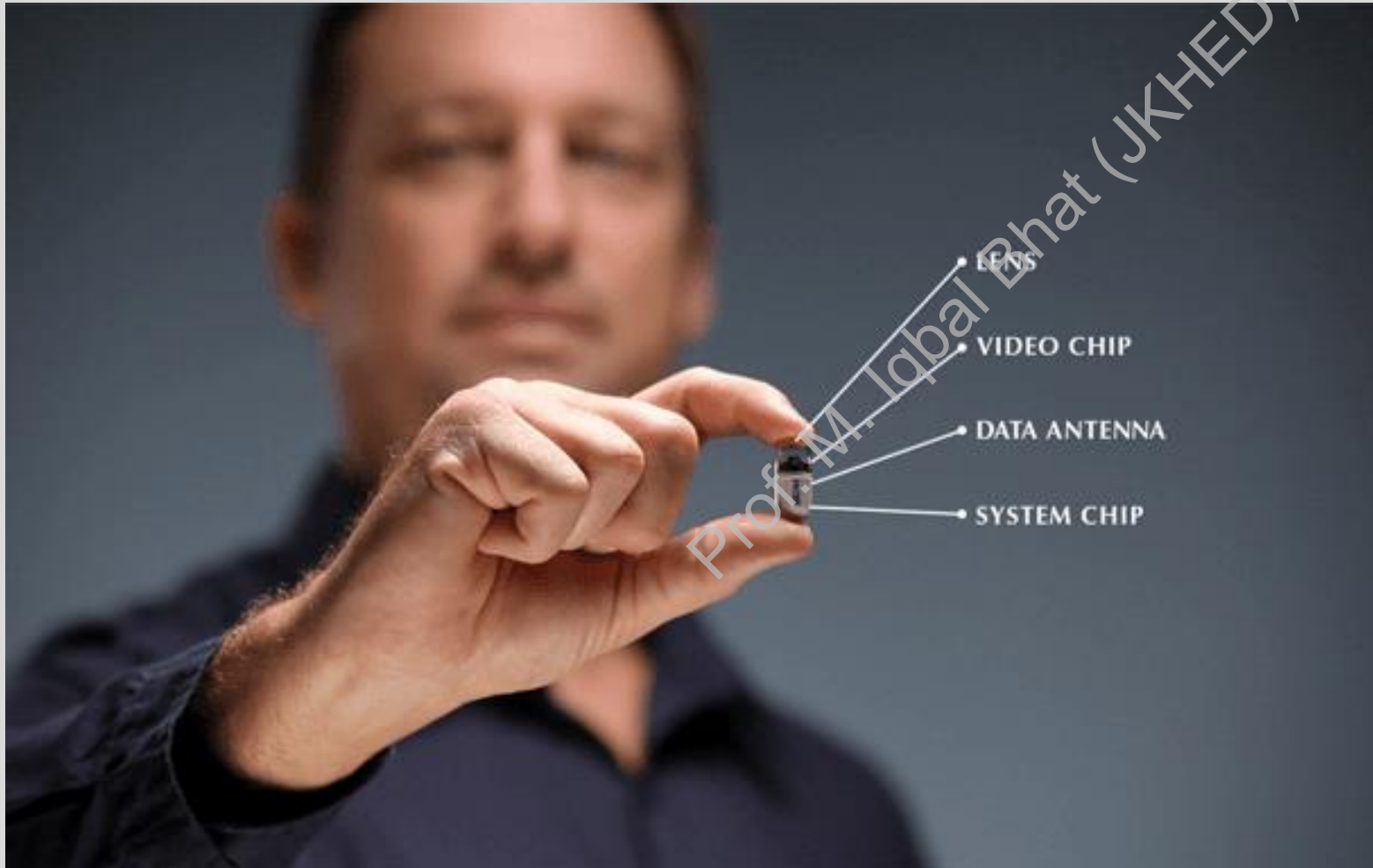
Activity trackers during cancer treatment



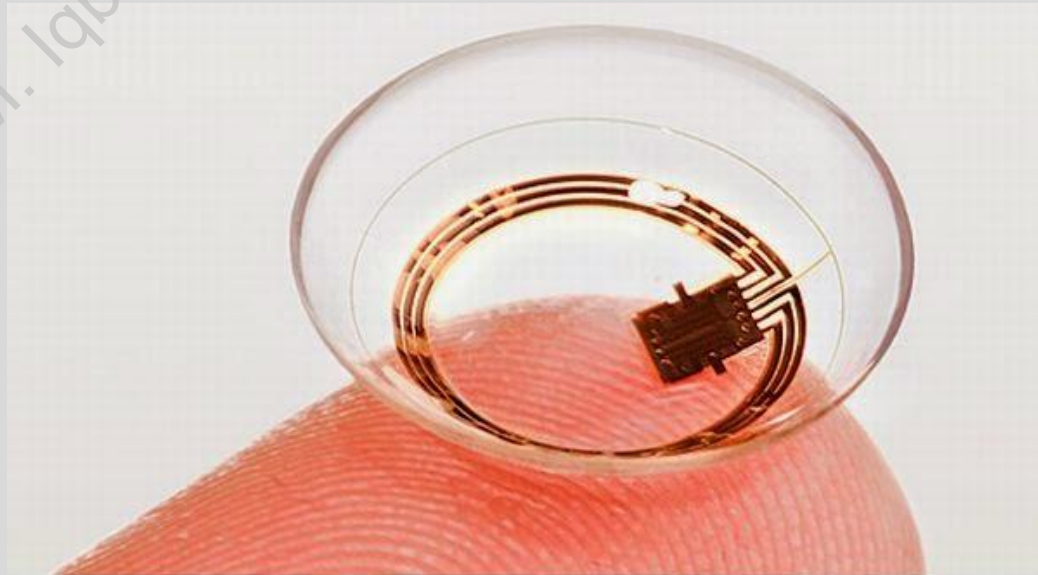
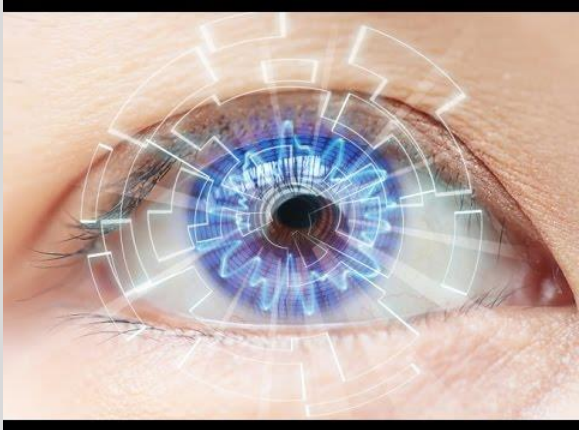
Connected inhalers



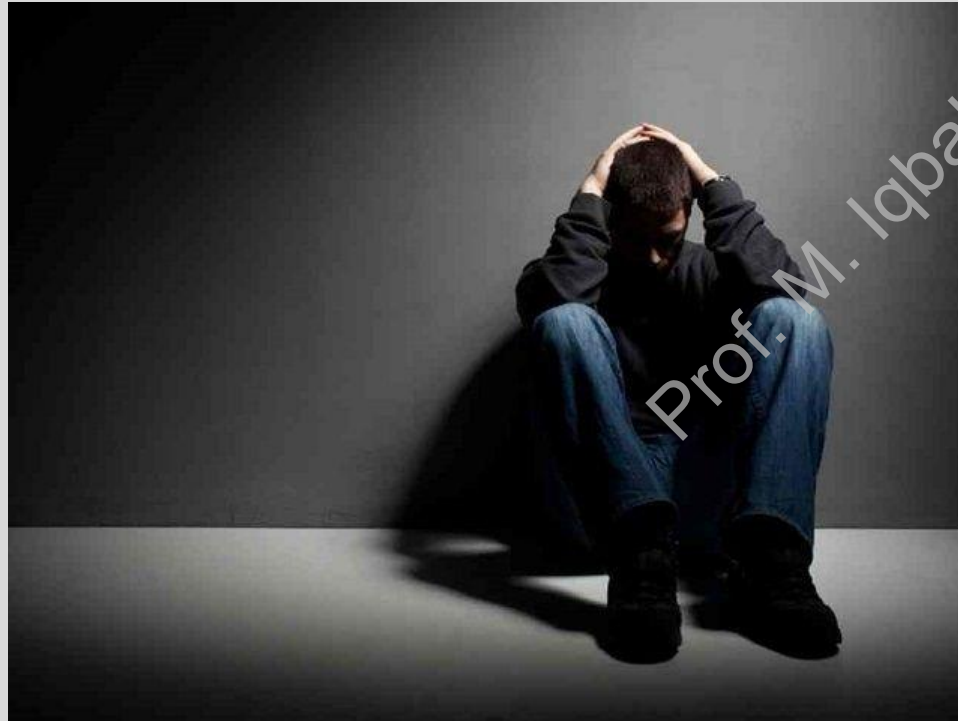
Ingestible sensors



Connected contact lenses



Depression-fighting Apple Watch app

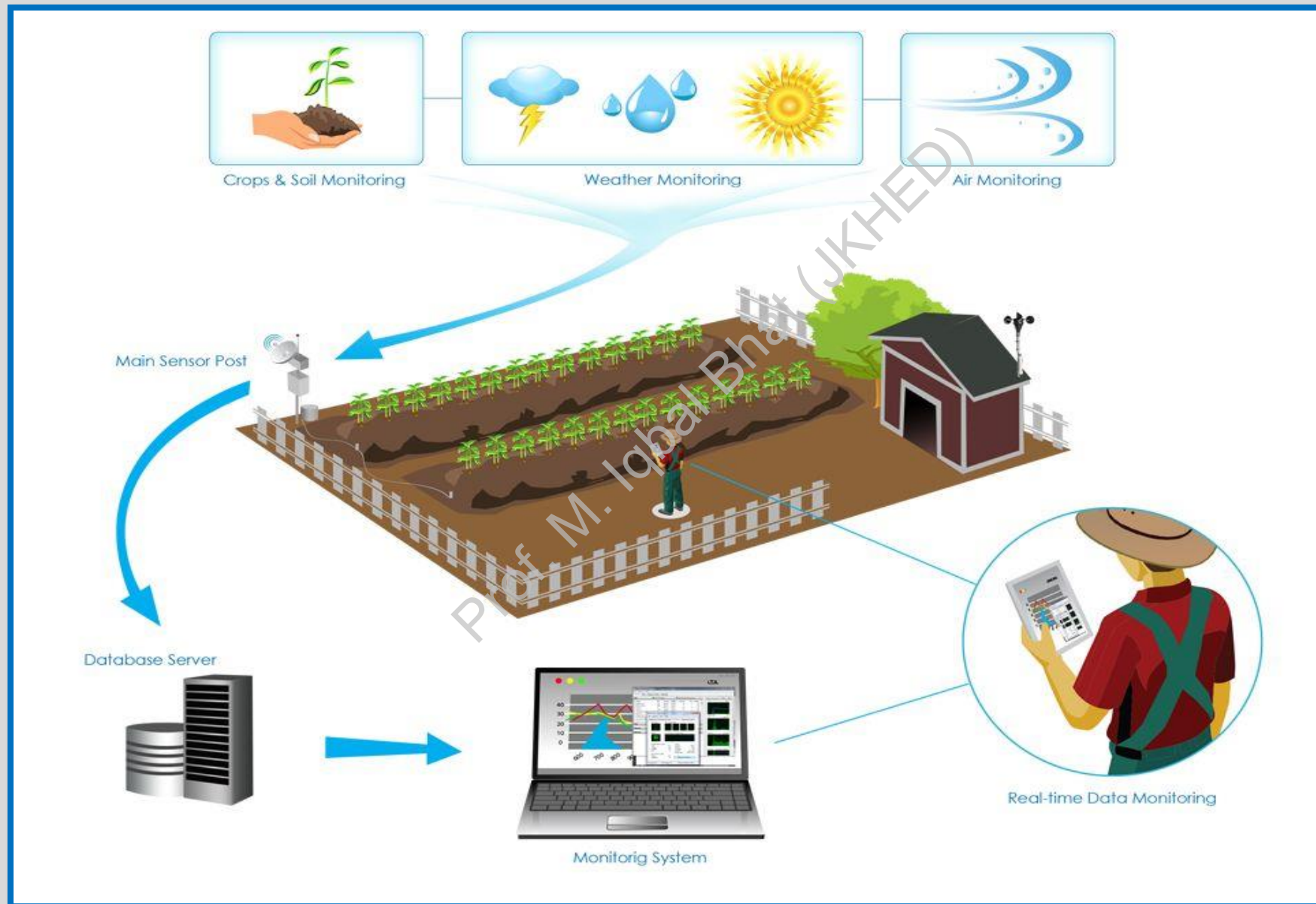


Other Applications

- ✓ Real Time Location Services
- ✓ Predicting the Arrival of Patients in PACU
- ✓ Hand Hygiene Compliance
- ✓ Tighten Budgets and Improve Patient Journey
- ✓ Remote Monitoring
- ✓ Focus on the Research Side of Healthcare



IoT in Agriculture



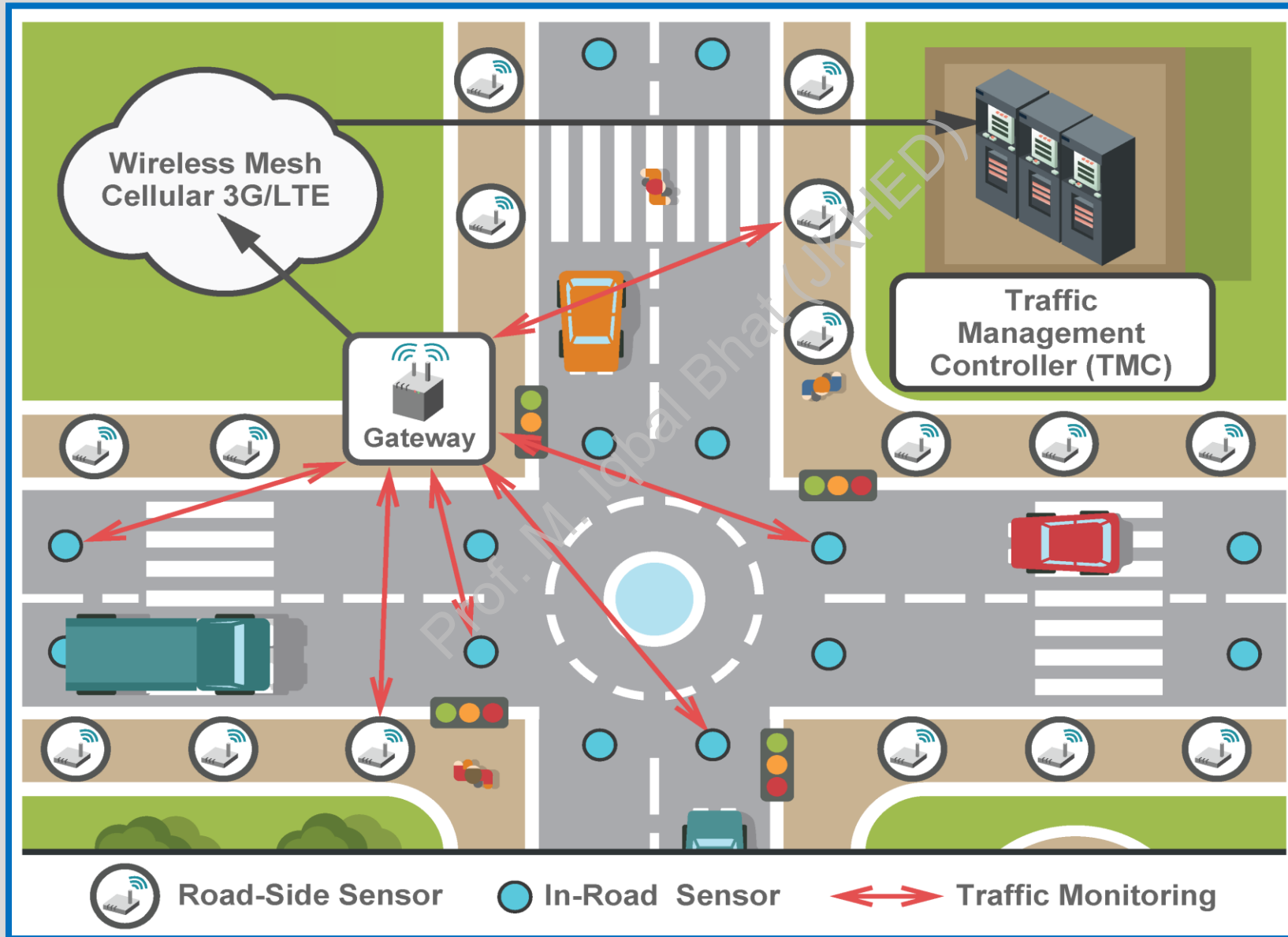
IoT in Agriculture



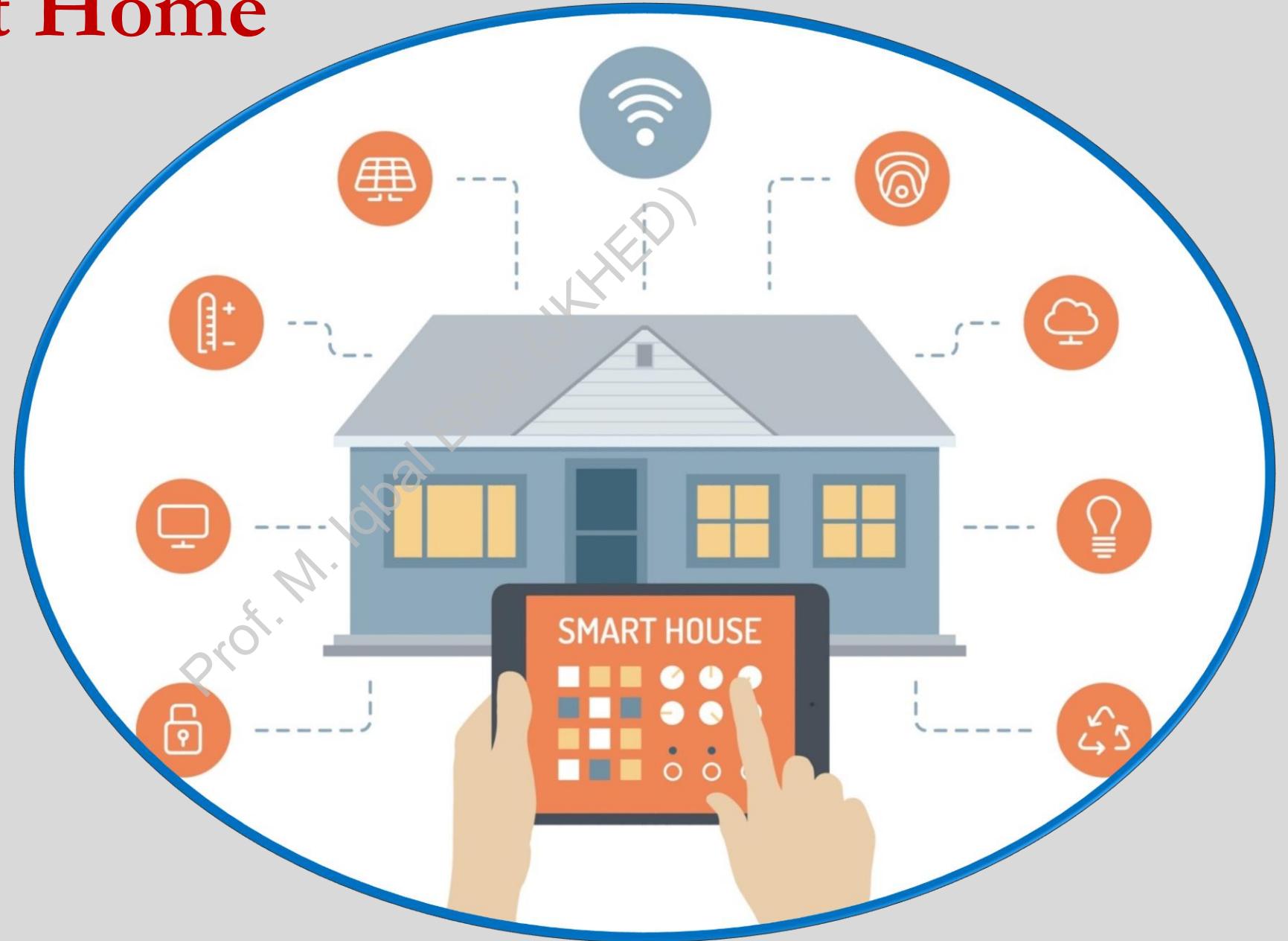
IoT in Traffic



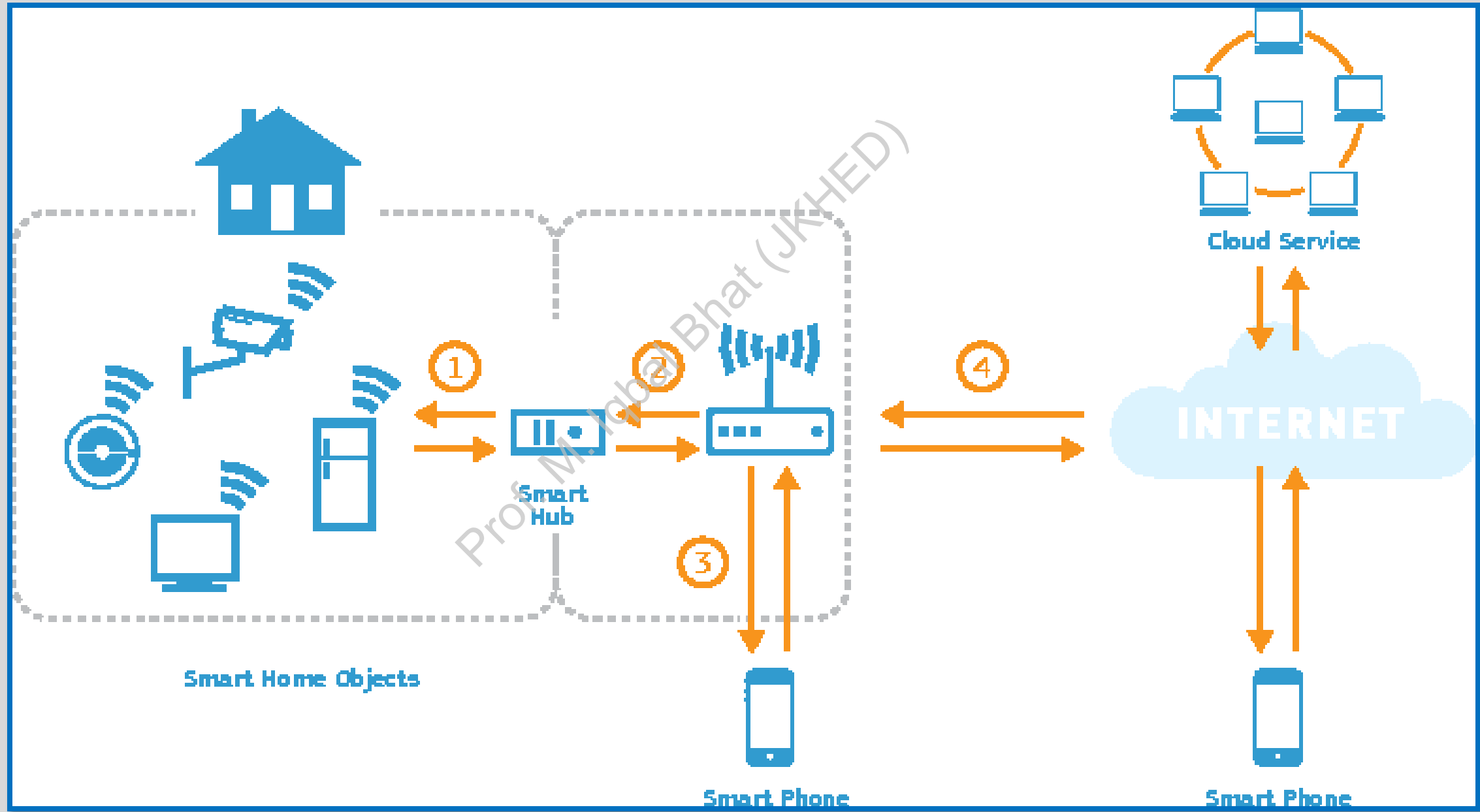
IoT in Traffic



IoT in Smart Home



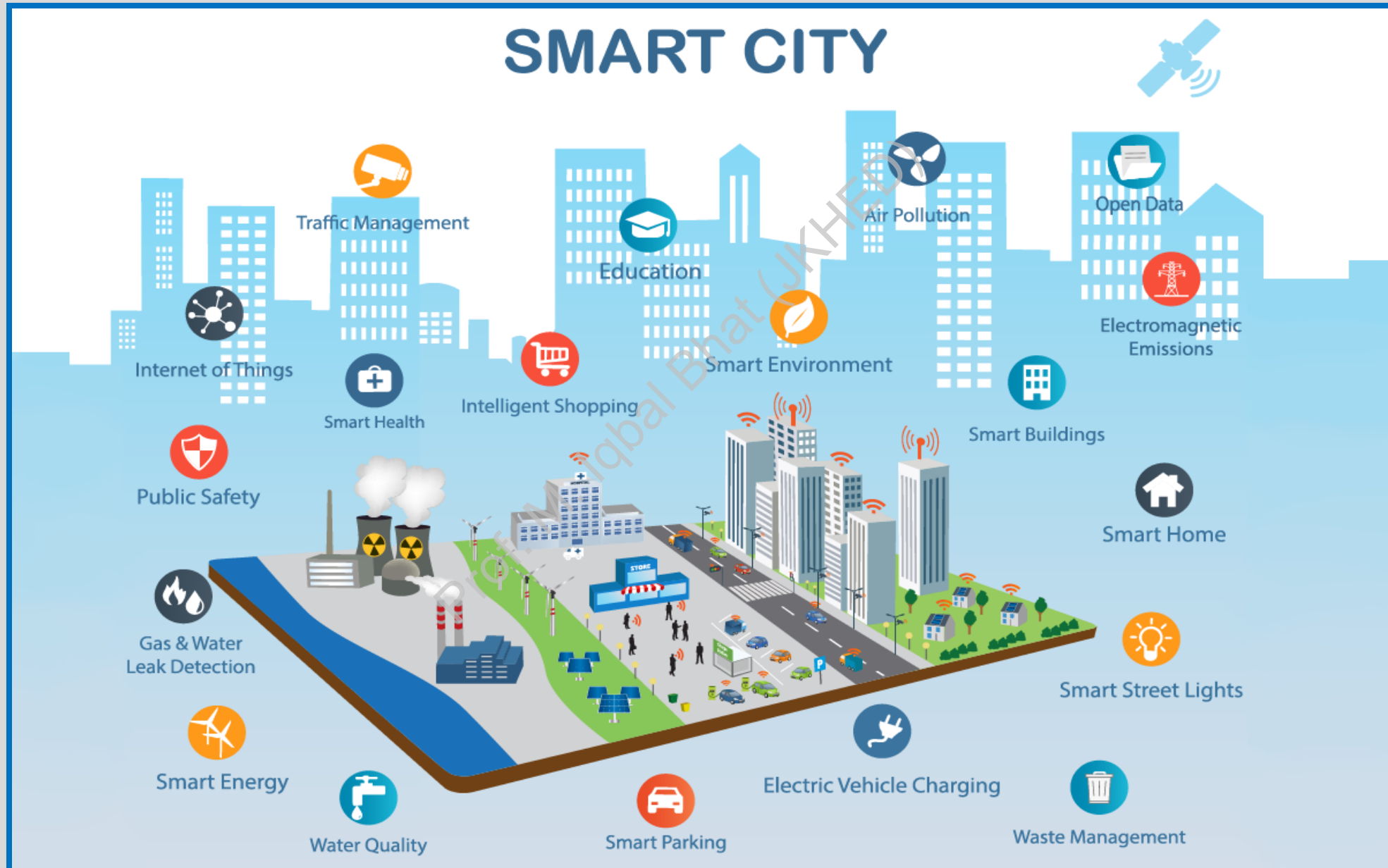
IoT in Smart Home



IoT in Retail



IoT in Smart Cities



IoT in Smart Cities

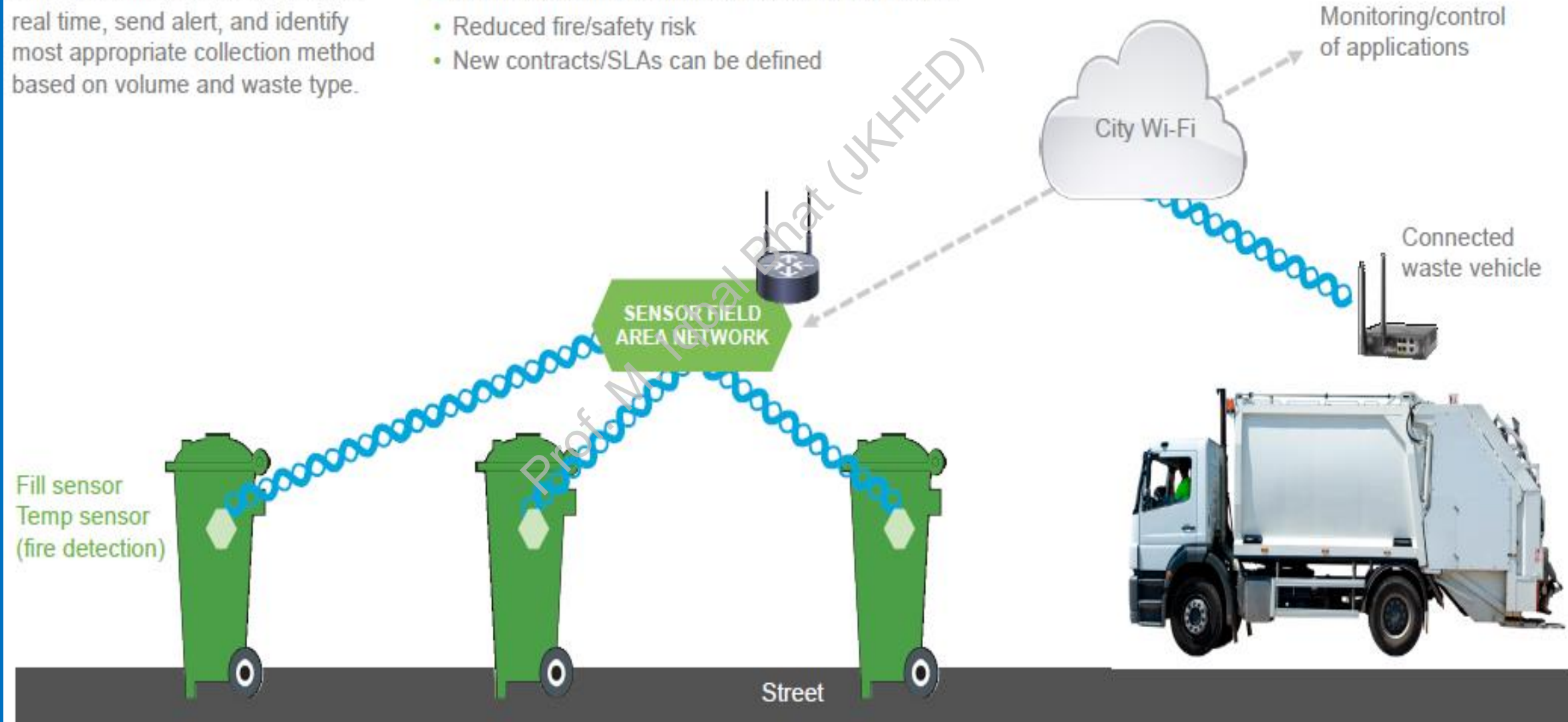


IoT Based Waste Collection

Sensors deployed in recycling containers monitor waste levels in real time, send alert, and identify most appropriate collection method based on volume and waste type.

Benefits include:

- Waste collection consumes less cost and carbon
- Reduced fire/safety risk
- New contracts/SLAs can be defined

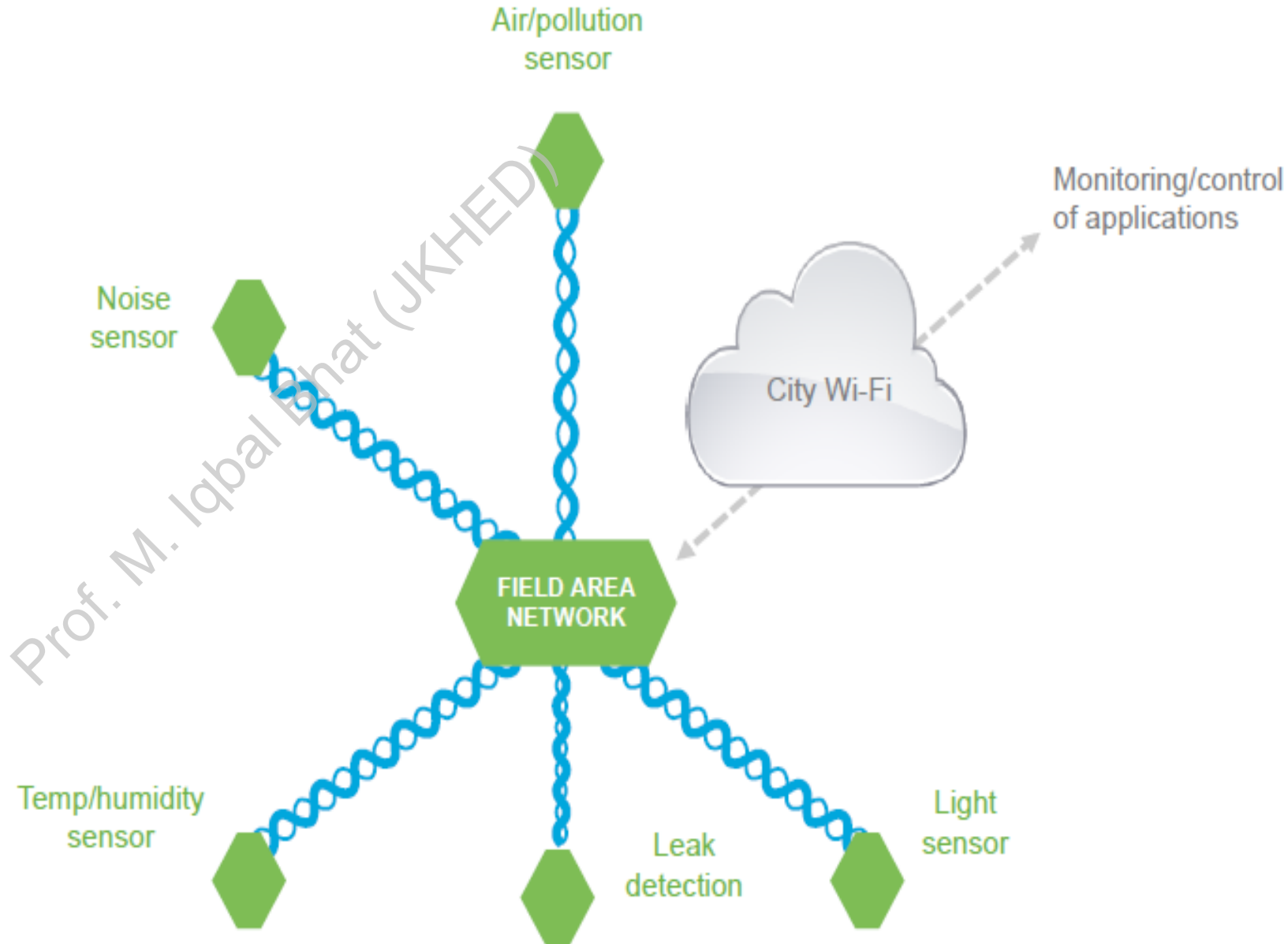


IoT Based Pollution Control

Installation of environment sensors:
air, light, humidity, noise, etc.

Benefits include:

- Leverages parking sensor infrastructure
- Provides valuable data for improving analytics applications and forecasting



IoT Applications are



ENDLESS